Zhiheng Zhang

Distributed AI infrastructure, consistency and fault-tolerant task execution

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Concordia application/student ID: 20814667 | Doctor of Philosophy - Computer Science

International applicant; neither a Canadian citizen nor a permanent resident.

Research profile

My research interest is the reliability of distributed agent execution: which updates may remain asynchronous, and which decisions need ordered, durable coordination? I bring systems architecture, AI-assisted integration and experiment/acceptance design, with a concrete two-worker software baseline rather than a claim of existing consensus-algorithm expertise.

Education

University of Melbourne, Australia
Master of Computer Science, 2025-present. Degree and final results pending. Master's research concerns Android privacy evidence; research supervisor: Professor Richard Sinnott.

University of British Columbia, Okanagan campus, Canada
Bachelor of Science in Computer Science, awarded 2024. Undergraduate study completed in English; capstone software engineering project: 93/100.

Selected research and systems experience

Utopia / Digital-City - multi-device systems and acceptance

- Led system design, AI-assisted integration and evidence-backed acceptance of Utopia, a completed and frozen integration of my DS-Hns and Codex Boss projects.
- Bounded physical evidence covers two Windows execution workers, Android/Web control, explicit target-device routing and cross-host result/replay records. Android is a control surface, not a validated third worker.
- Maintained task/evidence links and failure histories for reproducibility. The recorded REX-990 acceptance includes 205 reconstructed runs and 243 resolved trace pointers; these are scoped historical results, not a new test run or proof of general autonomy.
- Implementation stack: JavaScript ES modules/Node.js and Kotlin/Jetpack Compose. Digital-City holds architecture, task and acceptance records; it is not a separate deployed product.

Privacy Lens - Android privacy-evidence research

- Worked on an offline, evidence-first Android research prototype and a curated reproducibility artifact. The project separates admitted observations, technical signals and obligations requiring human review.
- Relevant practice includes versioned evidence admission, reproducible metric checks and claim-to-evidence traceability. The artifact includes TypeScript/React Native/Expo and a Kotlin native layer.
- The work does not establish legal compliance, clinical outcomes, independent detection accuracy or production readiness.

Target: Gengrui Zhang | Concordia Computer Science PhD

Earlier completed projects

- DS-Hns (completed/frozen): durable task state, restart/resume, recovery history and bounded code-task execution.
- Codex Boss (completed/frozen): multi-agent orchestration, inspectable completion evidence and acceptance governance. Both are predecessors of Utopia, not additional independent deployment claims.

Exploratory work and proposed research

- LucidGloves-derived ESP32 work: exploratory firmware and device-integration investigation, supporting an interest in embodied interaction. No validated custom glove or completed user study is claimed.
- Celestial Smart Glass: a documented research proposal for adaptive captions, session-level speaker association and user-controlled cues. No running prototype, measured hardware compatibility, participants or ethics approval is claimed.

Methods and technical practice

- Systems architecture, interface contracts and multi-device integration; designing bounded task lifecycles, state checks and recovery experiments.
- AI-assisted implementation with explicit review and acceptance; Git-based evidence/version management; test design, controlled fault injection, trace analysis and replay.
- Research emphasis: baselines and ablations, independently observable task outcomes, reproducibility and careful separation of simulation, physical observations and proposed work.

Targeted PhD direction

Can an agent task ledger coordinate only authorization-sensitive and externally consequential transitions while allowing ordinary progress updates to remain asynchronous, without increasing lost tasks, repeated effects or stale-permission execution under a stated failure model?

Proposed starting point: Separate speculative progress from committed external effects in a versioned ledger.

Selected relevant coursework

Analysis of Algorithms; Advanced Database Systems; Data Structures; Capstone Software Engineering Project.

Inspectable project evidence

Utopia - frozen implementation

REX-990 - bounded acceptance record

Privacy Lens - research artifact

Celestial Smart Glass - proposal only

Academic records accompany the email. Melbourne results are current and incomplete, not a final master's transcript. References can be arranged on request.